SINGLE-LAYER HYBRID MINIATURIZED FREQUENCY SELECTIVE RASORBER FOR SUB-TERAHERTZ APPLICATIONS

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Frequency Selective Rasorbers (FSRs) combine a bandpass frequency selective surface with a lossy absorbing layer, so that a structure passes an in-band signal with low insertion loss while absorbing out-of-band energy instead of reflecting it. This project presents a single-layer hybrid miniaturized FSR for sub-terahertz applications in the 0–2 THz range, designed and simulated in CST Studio Suite. The structure fits a 60 μm × 60 μm unit cell (about $0.2\lambda_0$ at 1 THz) on a silicon dioxide substrate with gold patterns. Miniaturization is obtained by path-length enhancement. A folded fractal bandpass layer, whose current path grows from 220 μm to 1217.6 μm, gives a 1 THz transmission resonance with a peak insertion loss of 0.87 dB. A convoluted square-spiral resistive layer with graphene on the feeder stem and third turn gives a continuous −10 dB reflection response from 0.678 to 1.377 THz (69.89% fractional bandwidth). Parametric studies of path length, track width, substrate thickness and graphene placement explain how each parameter shapes the response. The two layers are then integrated into the hybrid FSR, and the result is compared with reported terahertz FSRs. The design suits 6G communication, sensing, imaging and stealth radomes.

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LIST OF ABBREVIATIONS

A-T-A	Absorption–Transmission–Absorption
AT / TA	Absorption–Transmission / Transmission–Absorption
Au	Gold
CST	Computer Simulation Technology (CST Studio Suite)
EM	Electromagnetic
EMI	Electromagnetic Interference
FBW	Fractional Bandwidth
FSR	Frequency Selective Rasorber
FSS	Frequency Selective Surface
ISAC	Integrated Sensing and Communication
LC	Inductor–Capacitor (resonance)
PCB	Printed Circuit Board
RCS	Radar Cross Section
SiO_2	Silicon dioxide
SIW	Substrate Integrated Waveguide
THz	Terahertz
VO_2	Vanadium dioxide
2D / 2.5D / 3D	Two-, two-and-a-half- and three-dimensional
6G	Sixth-generation wireless network
λ_0	Free-space wavelength at the centre frequency
S11	Reflection coefficient (dB)
S21	Transmission coefficient (dB)
f_L, f_H, f_c	Lower, upper and centre frequency of a band
P_x, P_y	Unit cell periodicity along x and y

CHAPTER 1
INTRODUCTION TO TERAHERTZ FREQUENCY SELECTIVE RASORBERS (FSR)

1.1 BACKGROUND AND MOTIVATION

The terahertz (THz) band occupies the part of the electromagnetic spectrum between microwaves and infrared light, conventionally taken as 0.1–10 THz. Because it sits between electronics and photonics, it combines properties of both: radiation in this band is non-ionizing, can penetrate many non-polar and dielectric materials such as plastics, paper, textiles and ceramics, and offers an extremely wide usable bandwidth together with short wavelengths that allow fine spatial resolution. The portion of the band between 0.1 THz and 2 THz, referred to in this report as the sub-terahertz (sub-THz) range, is of particular interest because it is the region most actively considered for next-generation wireless links and for compact sensing and imaging systems.

Growing wireless traffic is already saturating the available GHz spectrum. The sub-THz band offers a large unused bandwidth, the possibility of data rates of the order of 100 Gbps, ultra-low latency and high-resolution sensing and imaging. These characteristics explain the interest of the research community in developing passive components, such as filters, absorbers and radomes, that can operate reliably in this range.

Why move from GHz to THz. Wireless communication has so far relied on the microwave and millimetre-wave (GHz) bands, which are shared by cellular, Wi-Fi, satellite, radar and broadcast services and are already heavily allocated. The data rate that a link can support grows with the bandwidth available to it, and in the GHz range only a few hundred megahertz to a few gigahertz of contiguous spectrum can usually be obtained. Better modulation and coding improve the spectral efficiency only to a limited extent, so GHz

systems cannot readily reach the multi-gigabit to terabit-per-second rates expected from 6G, high-resolution sensing and ultra-fast short-range links. At THz frequencies, tens to hundreds of gigahertz of contiguous bandwidth can be used, which makes the move from GHz to THz the most direct way of increasing capacity.

A shorter wavelength brings further advantages over the GHz range. At 1 THz the free-space wavelength is 300 μm, compared with 3 mm at 100 GHz and 30 mm at 10 GHz, so antennas, filters and radomes shrink to the micrometre scale and can be built as thin planar metasurfaces. Short wavelengths and narrow beams give finer spatial resolution for imaging and radar, and highly directional links reduce interference between nearby users, so that the same spectrum can be reused. The lower part of the THz band, up to about 2 THz, also still penetrates dielectric materials and suffers less atmospheric attenuation than the higher THz region, so it is the first part of the band considered for practical links.

The move to THz also brings difficulties: free-space path loss and atmospheric and material absorption are higher than in the GHz range, and components have to be very small and accurately fabricated. Passive structures designed specifically for this range, such as the frequency selective rasorber presented in this report, are therefore needed to control which signals are passed and which are absorbed in a THz front end.

1.1.1 Terahertz Communication (6G and Ultra-Wideband Systems)

Sixth-generation (6G) wireless networks are expected to support peak data rates far beyond those of current systems, together with very low latency and integrated sensing and communication (ISAC). Meeting these requirements needs contiguous bandwidth of tens of gigahertz, which is available only at sub-THz and THz frequencies. At these frequencies, wavelengths are of the order of a few hundred micrometres, so the antennas, filters and protective covers that

make up a front end are physically small, and passive spatial filters can be built as thin planar metasurfaces.

A THz communication link must, however, coexist with other emitters and with radar systems operating nearby. A radome that passes the desired channel with very low loss and absorbs everything else, instead of reflecting it, can therefore protect the receiver from out-of-band interference and at the same time reduce the radar signature of the platform that carries it. This is precisely the function of a frequency selective rasorber.

1.1.2 Terahertz Sensing and Non-Invasive Imaging

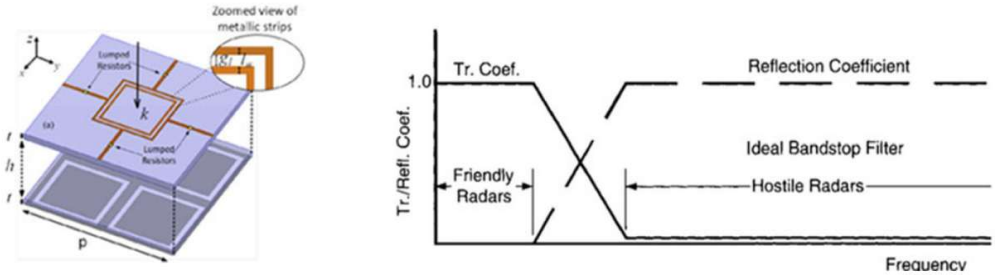
Terahertz radiation is strongly sensitive to the dielectric properties and molecular resonances of materials, while its photon energy is too low to cause ionization. This makes it attractive for non-destructive testing, security screening, spectroscopy, quality inspection and biomedical imaging, where the object under test must not be damaged. The short wavelength gives higher image resolution than microwaves, and many packaging and clothing materials are semi-transparent in this band. Sensing systems that operate in cluttered electromagnetic environments benefit from front-end structures that transmit only the band of interest and absorb unwanted signals, so that they do not generate spurious reflections or pick up interference.

1.2 ROLE OF FREQUENCY SELECTIVE RASORBERS IN SUB-THZ SYSTEMS

Radar systems detect targets by transmitting electromagnetic waves and receiving their reflections. This motivated the development of stealth technologies that reduce radar reflections and the radar cross section (RCS). Conventional electromagnetic absorbers, however, also block the signals needed for communication and sensing. A Frequency Selective Surface (FSS) can pass a chosen band, but outside that band it simply reflects the incident

wave, which increases bistatic RCS. A **Frequency Selective Rasorber (FSR)** resolves this trade-off by combining an FSS with an electromagnetic absorber: it transmits the desired frequencies with very low insertion loss, absorbs the unwanted frequencies instead of reflecting them, and reduces RCS while maintaining the communication capability. Unlike a conventional radome, which only protects an antenna, an FSR provides electromagnetic transparency and stealth functionality simultaneously. Typical applications are THz communication and sensing, stealth aircraft and naval platforms, radar communication systems, antenna radomes and electromagnetic interference (EMI) suppression.

When a plane wave strikes an FSR array, surface currents are induced on the resonator geometry. In the passband, the bandpass layer lets the wave through with minimal loss. In the stopband, the incident energy is dissipated as heat in resistive elements instead of being backscattered. The ideal spectral behaviour of an FSR is shown in Fig. 1.1.



(a) (b)

Fig. 1.1 (a) Frequency Selective Rasorber (b) Ideal Spectral Response of FSR

Spectral response modes

FSRs are classified by the relative positions of the absorption (A) and transmission (T) bands, as shown in Fig. 1.2.

- **Absorption–Transmission (A-T):** absorption occurs at lower frequencies and transmission at higher frequencies. A lossy layer is usually stacked above a bandpass spatial filter.

- **Transmission–Absorption (T-A):** the low-frequency band is a low-loss passband and higher frequencies are absorbed, which suppresses higher-frequency harmonics and secondary reflections.
- **Absorption–Transmission–Absorption (A-T-A):** two absorption bands surround one transmission band. This configuration absorbs out-of-band interference on both sides of the operating channel while keeping a lossless in-band communication window. It is therefore preferred for stealth radomes, 6G radomes, ISAC systems and multi-band THz receivers.

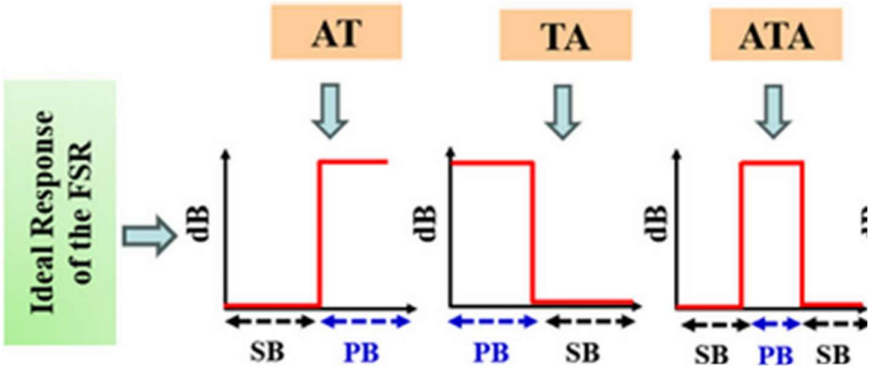


Fig. 1.2 Spectral Response Modes

Structural classification

FSRs are also classified by their physical structure (Fig. 1.3). Two-dimensional (2D) FSRs combine planar lossy layers and bandpass FSS arrays on planar substrates. They may be single-layer, dual-layer (with an air spacer or dielectric between the transmission and absorption layers, sometimes with lumped resistors or active elements) or multi-layer (stacked filters for steeper roll-off at the cost of thickness). Two-and-a-half-dimensional (2.5D) FSRs add vertical conductive paths such as via holes, which lengthen the current path inside a small footprint and improve miniaturization and angular stability. Three-dimensional (3D) FSRs use substrate-integrated waveguides, parallel-

plate waveguide arrays or magnetic material blocks to obtain high selectivity and wide-angle stability.

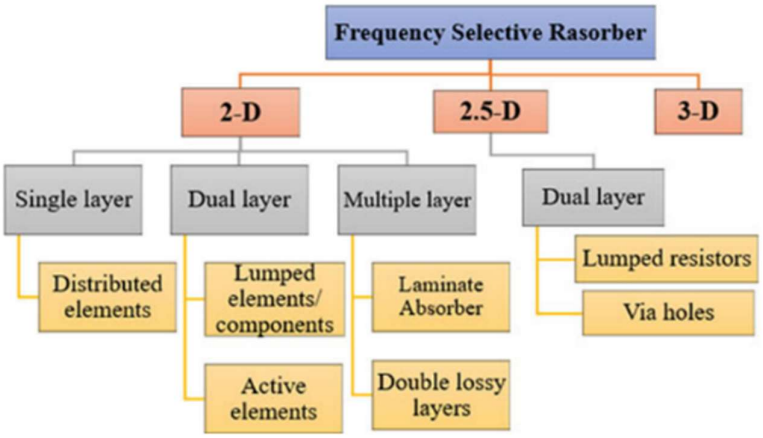


Fig. 1.3 Classification of Frequency Selective Rasorber

Single-layer FSRs place the absorbing and transmitting resonators on a single thin dielectric sheet. They offer a low profile and are suitable for conformal mounting, but the coplanar elements couple strongly and absorption bandwidth is usually narrow. The present work adopts a hybrid single-layer topology: a folded fractal bandpass pattern and a graphene-loaded spiral resistive pattern are realised on one silicon dioxide substrate, without an air spacer and without lumped resistors, which are difficult to fabricate at THz frequencies.

1.3 NEED FOR UNIT CELL MINIATURIZATION IN METASURFACE RADOMES

The unit cell of a metasurface radome should be small compared with the wavelength. A small cell keeps the response stable when the wave arrives obliquely, pushes grating lobes away from the operating band and allows the structure to be curved to follow the shape of an aircraft or antenna cover. A larger cell, on the other hand, limits angular stability. In this work, a conventional square-loop resonator with a 120 µm unit cell resonates near 0.64

THz and is therefore unsuitable for a compact radome. Reported unit cells in the literature range from about $0.20\lambda_0$ to well above one wavelength.

Miniaturization is obtained by lengthening the current path without enlarging the cell. A longer path raises the effective inductance, and the close spacing of folded segments raises the distributed capacitance, so the LC resonance moves to a lower frequency for the same footprint. In the transmission layer of this project, the current path grows from 220 μm to 1217.6 μm, roughly twenty times the side of the 60 μm cell, while the cell stays at about $0.2\lambda_0$ at the 1 THz design frequency. The absorption layer follows the same idea with convoluted square spirals whose track width and gap are reduced from 1 μm to 0.6 μm to accommodate more turns.

1.4 ORGANIZATION OF THE REPORT

The report is organized into five chapters.

- **Chapter 1** introduces the background and motivation for moving from the GHz to the terahertz range, terahertz communication and sensing applications, the role of frequency selective rasorbers in sub-THz systems and the need for unit cell miniaturization.
- **Chapter 2** reviews existing terahertz FSRs and metamaterial absorbers, the materials and designs they use and how they obtain their results, compares them, and derives the inferences, research gap and problem statement.
- **Chapter 3** describes the design and methodology: specifications, materials, the stage-wise evolution of the transmission layer and absorption layer, and their integration into the single-layer hybrid FSR.
- **Chapter 4** presents and discusses the simulated results, including the effect of graphene loading and its placement, the fractional bandwidth of

the transmission and absorption layers, a comparison with the literature, and the merits of the proposed design.

- **Chapter 5** concludes the work, lists its contributions and limitations and outlines future scope.

CHAPTER 2
LITERATURE REVIEW ON FREQUENCY SELECTIVE RASORBERS

2.1 OVERVIEW OF EXISTING TERAHERTZ FSR TECHNOLOGIES

Several studies have explored frequency selective rasorbers and terahertz metamaterial absorbers [1]–[7]. This chapter looks at how each design chooses its materials, how its geometry is arranged, and how the reported transmission and absorption are obtained. Most of the designs use a thin polyimide or similar substrate, because a thin, low-loss dielectric keeps the dielectric loss and the thickness small at terahertz frequencies, and gold metallization, because its high conductivity gives low-loss resonators for the transmission band. The absorption function is obtained from a lossy element, either a resistive film or vanadium dioxide (VO_2), which dissipates the incident energy outside the transmission band, and none of the designs uses lumped elements. In every design the transmission band comes from a resonant bandpass pattern and the absorption band from the lossy layer. The prototypes of the six reviewed designs are shown in Fig. 2.1, and their key parameters are listed in Table 2.1.

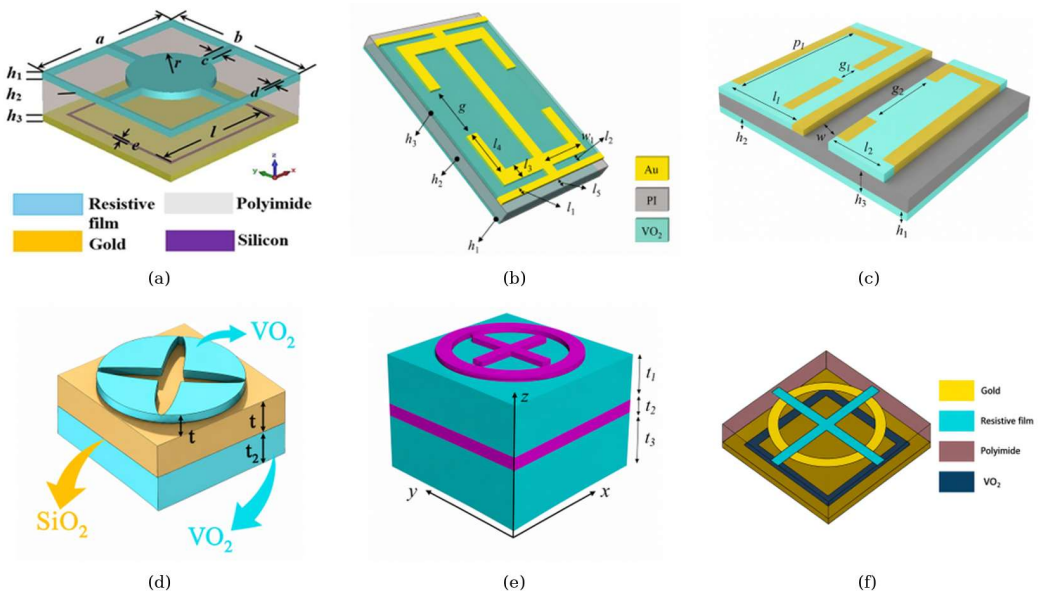


Fig. 2.1 Prototypes of the reviewed terahertz FSRs and switchable absorbers (a) Optically reconfigurable FSR/absorber (2025) (b) VO_2 device, absorption–transmission (2024) (c) VO_2 device, transmission–absorption (2022) (d) THz smart switch (2024) (e) Bifunctional metamaterial (2024) (f) Thermally controlled absorber (2023)

Table 2.1 Summary of reviewed terahertz FSRs and switchable metamaterial absorbers

Ref.	Year	Periodicity (µm)	Substrate / Metal	Transmission	Insertion loss	Absorption band
[3]	2025	20 × 20	Polyimide (1 µm) / Gold	3 THz	2.6 dB	1–10 THz
[7]	2024	160 × 100	Polyimide (25 µm) / Gold	0.486 and 1.053 THz	≤ 0.46 dB	0.44, 1.25, 1.93 and 2.45 THz
[6]	2022	150 × 150	Polyimide (20 µm) / Gold	0.90 THz	0.87 dB	0.66, 1.22, 2.14 and 2.48 THz
[5]	2024	35 × 35	SiO_2 (75 µm) / VO_2	5.3–10 THz	0.65 dB	3.6–9.4 THz
[2]	2024	24 × 24	Topas (9 µm) / VO_2	1–10 THz	< 0.46 dB	2.54–7.65 THz
[4]	2023	14 × 14	Polyimide (2 µm) / Gold	5.15 THz	4.29 dB	3.31–10 THz

2.2 REVIEW OF COMPACT POLYIMIDE-BASED TERAHERTZ DESIGNS

The wideband FSR/absorber reported in 2025 [3] uses a 1 µm polyimide substrate with gold patterns. Polyimide is a thin, low-loss film at terahertz

frequencies and gold gives low-loss bandpass resonators, so the 20 μm × 20 μm cell, the smallest of the reviewed designs with the exception of [4], is kept thin and compact. The absorption comes mainly from a resistive film separated from the bandpass pattern by a 75–100 μm air space, which corresponds to roughly a quarter wavelength at the lower end of the 1–10 THz absorption band, the usual spacing for resistive-sheet absorbers. This arrangement gives an A-T-A response: the resistive film and air space absorb from 1 THz to 10 THz, while the bandpass pattern transmits at 3 THz with an insertion loss of 2.6 dB, suitable for THz radar, radomes and RCS reduction. A silicon layer is also included in the prototype for optical control. The 2.6 dB insertion loss is relatively high for a communication window and indicates the trade-off between a very small cell and low in-band loss.

The thermally controlled absorber reported in 2023 [4] uses a 14 μm × 14 μm cell on a 2 μm polyimide substrate with gold, a resistive film and VO_2. The thin polyimide and gold are used for the same reasons as in [3], while VO_2 is used because its conductivity rises by several orders of magnitude above its phase-transition temperature, so that it behaves as a lossy metal in one state and as a dielectric in the other. In the lossy state it works with the resistive film to give wideband absorption from 3.31 THz to 10 THz, and in the dielectric state the gold pattern transmits at 5.15 THz. The insertion loss of 4.29 dB is the highest of the reviewed designs, and the application areas are radar stealth and selective transceivers. Both [3] and [4] transmit at 3 THz or higher and have the smallest cells of the review (14–20 μm), and both pay for this compactness with a higher insertion loss.

2.3 REVIEW OF VO_2-BASED TERAHERTZ DESIGNS

Vanadium dioxide changes from an insulating to a metallic state with temperature, which is why it appears in [2], [5], [6] and [7]: the same pattern is nearly transparent when VO_2 is insulating and lossy when it is metallic, so a

single structure provides both the transmission and the absorption function without lumped resistors. The device in [7], reported in 2024, uses a 160 μm × 100 μm cell on a 25 μm polyimide substrate with gold and obtains transmission at 0.486 THz and 1.053 THz with an insertion loss up to 0.46 dB, and absorption at 0.44, 1.25, 1.93 and 2.45 THz; its applications are THz imaging, sensing and wireless communication. The earlier device in [6] (2022) uses a 150 μm cell and a 20 μm polyimide substrate, transmits at 0.90 THz with 0.87 dB insertion loss and absorbs at 0.66, 1.22, 2.14 and 2.48 THz, for THz switching, sensing and security. The large cells of 150–160 μm are what allow these two designs to resonate below 2.5 THz, and the several transmission and absorption frequencies are consistent with the multiple resonant modes of such a large cell; the price is an electrical size of $0.45\lambda_0$–$0.56\lambda_0$ (Table 4.4).

The THz device in [5] uses a 35 μm cell on a 75 μm SiO_2 substrate, a stable, low-loss dielectric widely used in thin-film processing. It transmits over 5.3–10 THz with 0.65 dB insertion loss and absorbs over 3.6–9.4 THz, and is aimed at smart windows, non-destructive testing and THz imaging. The bifunctional design in [2] uses a 24 μm cell and a 9 μm Topas substrate, a polymer with low loss and low dispersion at THz frequencies; it transmits over 1–10 THz with insertion loss below 0.46 dB and absorbs over 2.54–7.65 THz, for THz modulation, stealth and broadband absorbers. In these VO_2 devices the transmission and absorption ranges refer to the two states of the VO_2, and the wide bands in [2] and [5] are obtained by combining several closely spaced resonances over a broad range.

A broader meta-analysis of frequency selective rasorbers [1] summarizes the 2D, 2.5D and 3D design families, the use of lumped components and the trade-offs between thickness, bandwidth and angular stability, and provides the classification used in Chapter 1.

Graphene has also been used as the lossy material in terahertz rasorbers and absorbers with a transmission band [8]–[10], because it replaces the lumped resistors that are too large for terahertz unit cells and its conductivity can be tuned through its chemical potential. The rasorber in [8] cascades a lossy graphene layer, a dielectric spacer and a lossless bottom layer containing a complementary square ring resonator, and obtains absorption from 1.27 THz to 2.43 THz with a low insertion loss at 2.8 THz. The absorber in [9] places patterned graphene on a SiO_2 substrate with a frequency selective surface (FSS) on the bottom, which gives two passbands, one below 0.19 THz and one near 1.49 THz, while the FSS acts as a ground plane in the absorption bands. The design in [10] uses a sinusoidal graphene layer on SiO_2 backed by a bandpass FSS, which gives a broad absorbing band from 0.5 THz to 1 THz and a transmission window outside it. These works support the use of graphene on SiO_2 as the lossy material, as adopted in this work.

2.4 SUMMARY AND COMPARATIVE EVALUATION OF REVIEWED LITERATURE

Table 2.1 shows that the reviewed devices share several features. All of them obtain absorption from a lossy layer (VO_2, silicon or a resistive film) combined with a resonant gold or VO_2 pattern, all use thin metal–dielectric stacks with no lumped elements, and most have unit cells between 14 μm and 160 μm. Insertion losses range from below 0.46 dB to 4.29 dB, and the two smallest cells, [4] (14 μm) and [3] (20 μm), have the highest losses, 4.29 dB and 2.6 dB. Their operating frequencies are mostly in the 2.5–10 THz region, with only the designs of [6] and [7] working below about 2.5 THz, and these have comparatively large cells of 150–160 μm. A compact, passive FSR for the 0–2 THz range, with a stable transmission window and a wide absorption band in the same band of operation, is therefore not addressed by the reported work.

2.5 INFERENCES FROM LITERATURE REVIEW

- **Limited focus on the 6G sub-THz band.** Most reported THz FSRs operate in the 2.5–10 THz range. Very few designs specifically target the 0–2 THz spectrum required for 6G communications and THz sensing.
- **Miniaturization remains a challenge.** The reported unit cells have electrical lengths from about $0.20\lambda_0$ to $1.30\lambda_0$. Achieving sub-wavelength ($< 0.3\lambda_0$) compact FSRs at sub-THz frequencies remains difficult.
- **Reliance on additional functional materials.** Many of the designs use VO_2 or silicon, which need an external thermal or optical stimulus to work as intended. A fully passive design built from SiO_2, gold and graphene is simpler to realise and operate.
- **Iterative design methodology.** Existing works rely on repeated parametric optimization in full-wave tools such as CST, which is time-consuming for multi-objective FSR design. A systematic, stage-wise geometric evolution that links each modification to a physical effect makes the design process traceable.

2.6 RESEARCH GAP AND PROBLEM STATEMENT

Research gap. There is a need for a compact, passive FSR operating in the 0–2 THz band that provides high in-band transmission and wideband out-of-band absorption, suitable for 6G and sensing applications, with a unit cell well below $0.3\lambda_0$.

Problem statement. Growing wireless traffic is saturating the available GHz spectrum, and the sub-THz band offers large unused bandwidth. To use it, a front-end structure must transmit the desired band with very low insertion loss and absorb unwanted signals, so that efficient communication with EMI suppression is possible. The problem addressed in this project is to design a single-layer hybrid miniaturized FSR, with a 60 μm × 60 μm unit cell, that

provides a transmission band around 1 THz and a wideband absorption response in the 0–2 THz range, using only passive materials (silicon dioxide, gold and graphene).

CHAPTER 3
DESIGN AND METHODOLOGY

3.1 DESIGN GOALS AND SPECIFICATIONS

The main goal of this project is to design a miniaturized single-layer hybrid FSR operating in the 0–2 THz frequency range for 6G communication and terahertz sensing applications. The design is carried out in CST Studio Suite on a 60 μm × 60 μm unit cell, a silicon dioxide (SiO_2) substrate and a gold conductive layer. The transmission layer is designed for an initial centre frequency of 1 THz, and the unit cell size of 60 μm corresponds to $0.2\lambda_0$ at that frequency. The absorption layer is designed to give a broad low-reflection (S11 ≤ -10 dB) response around the same band. Performance is evaluated using the reflection coefficient (S11) and the transmission coefficient (S21) obtained with periodic unit-cell boundary conditions in the x and y directions.

Choice of the frequency range. The upper limit of 2 THz and the 1 THz centre frequency were chosen for four reasons. First, the 0.1–2 THz region is the sub-THz range that is considered first for 6G links, sensing and imaging, and the review in Chapter 2 showed that the reported FSRs work mainly above 2.5 THz, so this range is not covered by compact reported designs. Second, atmospheric attenuation grows with frequency, so the lower part of the terahertz band is the most practical for communication links. Third, a 1 THz centre frequency inside a 0–2 THz span leaves room for an absorption band on each side of the transmission window, which gives the A-T-A arrangement described in Chapter 1, and the free-space wavelength of 300 μm at 1 THz means that a 60 μm cell is $0.2\lambda_0$, below the $0.3\lambda_0$ target for a sub-wavelength metasurface. Fourth, the range is realistic for fabrication and for the materials: the smallest features of the design (0.2–1 μm) are within the reach of electron-beam

lithography, and the behaviour of gold and graphene at these frequencies is well established.

Stability of gold. Gold is chosen as the conductor mainly because of its stability. It is chemically inert and does not oxidize in air, so its conductivity and the dimensions of the narrow tracks stay constant with time, which matters for tracks only 0.2–1 μm wide. At 1 THz the skin depth of gold is about 0.08 μm, much smaller than the 0.5 μm thickness used, so the patterns behave as good conductors with low loss and the resonance is not sensitive to small variations of the film thickness. Gold is also easy to pattern with standard lithography on oxide substrates. The reasons for choosing the SiO_2 substrate and the graphene loading are given in Section 3.2. The specifications are given in Table 3.1, and the design flow is shown in Fig. 3.1.

Table 3.1 Design specifications of the proposed FSR

Parameter	Specification
Operating frequency range	0–2 THz
Centre design frequency (initial)	1 THz
Unit cell size (periodicity)	60 μm × 60 μm ($0.2\lambda_0$ at 1 THz)
Substrate material	Silicon dioxide (SiO_2)
Substrate thickness	4 μm (transmission layer); 30 μm (absorption layer and integrated FSR)
Conductive layer	Gold, 0.5 μm thick
Lossy material	Graphene, 0.001 μm thick
Boundary conditions	Unit cell (periodic) in x and y directions
Performance parameters	S11 and S21

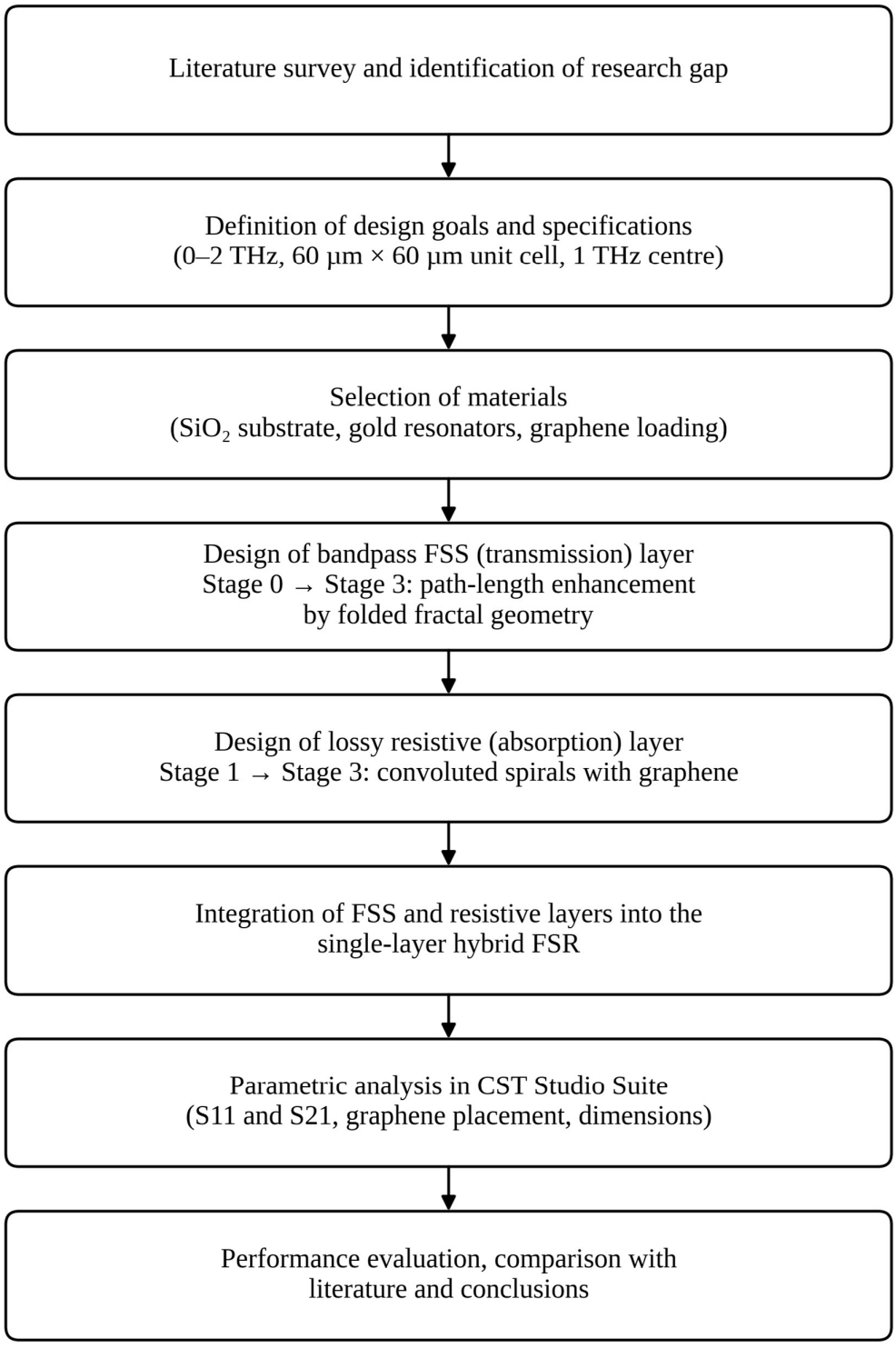


Fig. 3.1 Design flow of the single-layer hybrid miniaturized FSR

3.2 SELECTION OF MATERIALS AND DIELECTRIC SUBSTRATES

Substrate. Silicon dioxide is preferred as the substrate because of its electrical insulation, dielectric stability and suitability for terahertz applications, and because it is compatible with thin-film processing. The substrate thickness is one of the parameters tuned during the design: 11.5 μm is used in the early transmission-layer stages, 4 μm in the final transmission layer, and 30 μm in the absorption layer and the integrated structure.

Conductor. Gold is used for the resonator patterns because of its high electrical conductivity, chemical stability and dependable behaviour at THz frequencies. A thickness of 0.5 μm is used throughout.

Lossy material. Lumped resistors are difficult to integrate and fabricate reliably at THz frequencies because of parasitic effects and fabrication limits. Graphene is therefore used as an alternative lossy material. Its conductivity can be tuned to control electromagnetic absorption, and it is compatible with thin-film fabrication. In the design a 0.001 μm graphene layer is placed on selected parts of the spiral to add resistive loss and reactive loading. The materials are summarized in Table 3.2.

Table 3.2 Materials used in the proposed FSR and their role

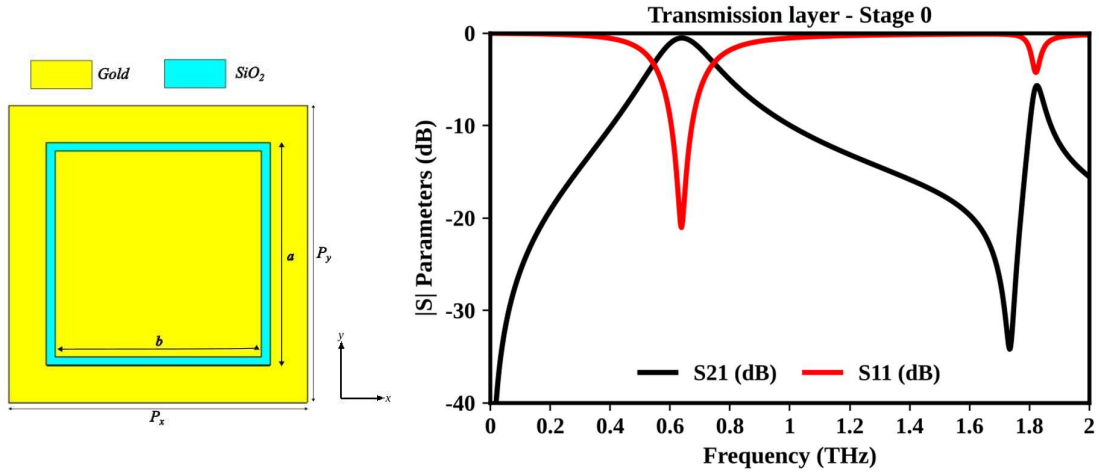
Material	Role in the design	Thickness
Silicon dioxide (SiO_2)	Dielectric substrate	4 μm to 30 μm (stage dependent)
Gold (Au)	Bandpass fractal resonator and spiral resonator patterns	0.5 μm
Graphene	Resistive loss and reactive loading on feeder stem and third spiral turn	0.001 μm

3.3 DEVELOPMENT AND EVOLUTION OF TRANSMISSION LAYER (FSS)

The bandpass FSS layer passes the desired operating frequency with minimum loss while controlling unwanted frequencies. It is designed for an initial target of 1 THz in a 60 µm cell, with gold patterns on a SiO_2 substrate. The resonator is evolved in four stages, and in each stage the effective current path is changed to move the resonance. The transmission layer is evaluated with S21; S11 shows how well the passband is matched.

3.3.1 Stage 0: Conventional Square Loop Resonator

The first transmission layer is a conventional square-loop resonator in a 120 µm unit cell with a current path of 360 µm, an SiO_2 thickness of 11.5 µm and a gold thickness of 0.5 µm (Fig. 3.2). The simulated response has a transmission resonance at about 0.64 THz, with a peak S21 of −0.51 dB and an S11 minimum of −21.0 dB. This confirms that the basic loop supports the required LC resonance. However, the large unit cell limits angular stability and compactness, so the resonator is compacted and folded to fit a 60 µm cell.

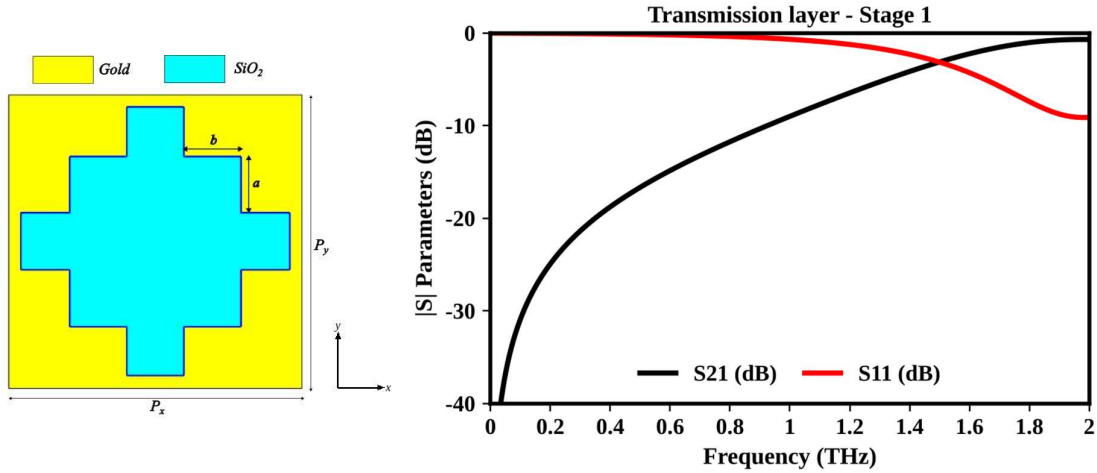


(a) **(b)**

Fig. 3.2 Structure of the FSS Bandpass Layer Stage-0 (a) Front View ($P_x = P_y$ = 120µm, a = 90µm, b = 83µm) (b) Simulated S-parameter characteristics

3.3.2 Stage 1: Simple Cross Resonator Structure

In Stage 1 the cell is reduced to 60 µm and a simple cross resonator is used, with a current path of 220 µm (Fig. 3.3). The simulated S21 rises almost monotonically with frequency and shows no distinct passband near 1 THz. The current path and the effective LC are too small to support the desired resonance, so the path length must be increased by fractal folding while keeping the same cell size.

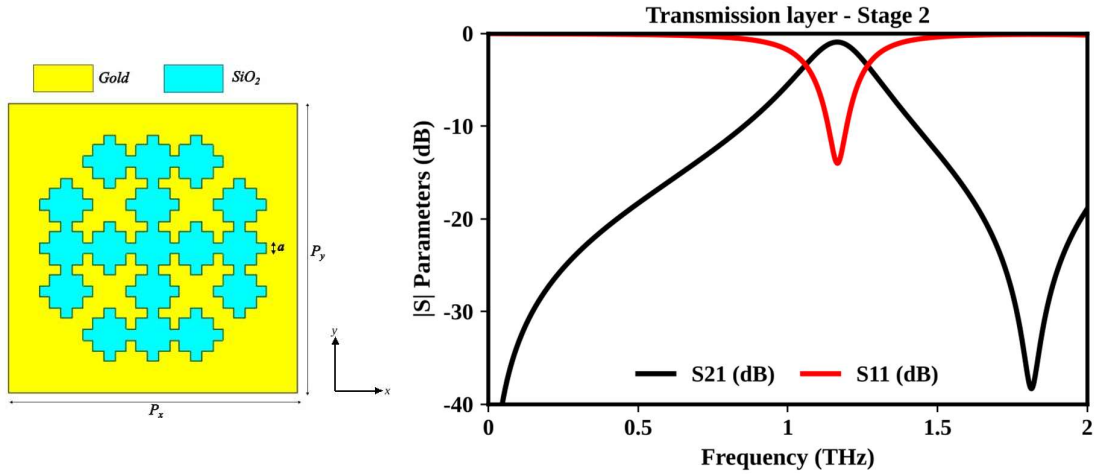


(a) **(b)**

Fig. 3.3 Structure of the FSS Bandpass Layer Stage-1 (a) Front View ($P_x = P_y = 60\mu m$, a = 11.667µm, b = 11.6665µm) (b) Simulated S-parameter characteristics

3.3.3 Stage 2: Folded Fractal Resonator Geometry

In Stage 2 the cross is replaced by a folded fractal resonator, which increases the current path to 609.331 µm in the same 60 µm cell (Fig. 3.4). The longer path adds distributed inductance and capacitance and produces a defined transmission passband centred at about 1.17 THz, with a peak S21 of −0.92 dB and an S11 minimum of −14.0 dB. This validates the folded fractal approach: increasing the path length lowers the resonant frequency while the unit cell is unchanged.

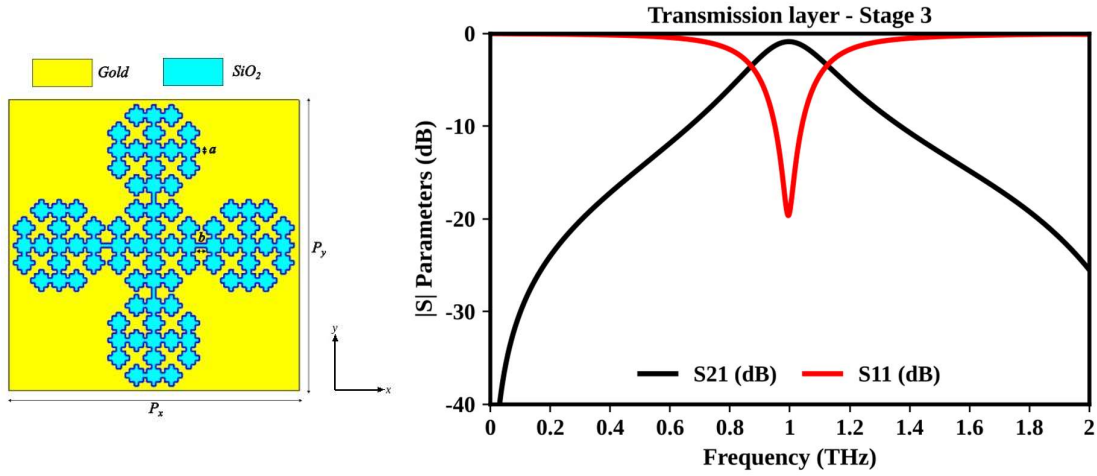


(a) **(b)**

Fig. 3.4 Structure of the FSS Bandpass Layer Stage-2 (a) Front View ($P_x = P_y = 60\mu m$, $a = 2.3334\mu m$) (b) Simulated S-parameter characteristics

3.3.4 Stage 3: Compacted Folded Fractal Structure (Target 1 THz)

In Stage 3 the fractal is folded further, giving a current path of 1217.6 µm in the 60 µm cell, and the SiO_2 thickness is reduced to 4 µm (Fig. 3.5). The folded geometry increases current confinement and inter-element capacitive coupling. The transmission resonance moves to the 1 THz target (peak S21 of −0.87 dB near 0.996 THz) and S11 reaches a minimum of −19.6 dB. The S11 ≤ −10 dB region spans about 0.973–1.021 THz. This shows electrical miniaturization: 1 THz resonance is obtained without enlarging the unit cell.



(a) **(b)**

Fig. 3.5 Structure of the FSS Bandpass Layer Stage-3 (a) Front View ($P_x = P_y = 60\mu m$, a = 0.93336µm, b = 2.4µm) (b) Simulated S-parameter characteristics

Evolution of the transmission layer. Fig. 3.6 compares S21 of Stages 1, 2 and 3, and Table 3.3 lists the parameters of all four stages. The fractional bandwidth of the transmission layer is evaluated in Section 4.2.

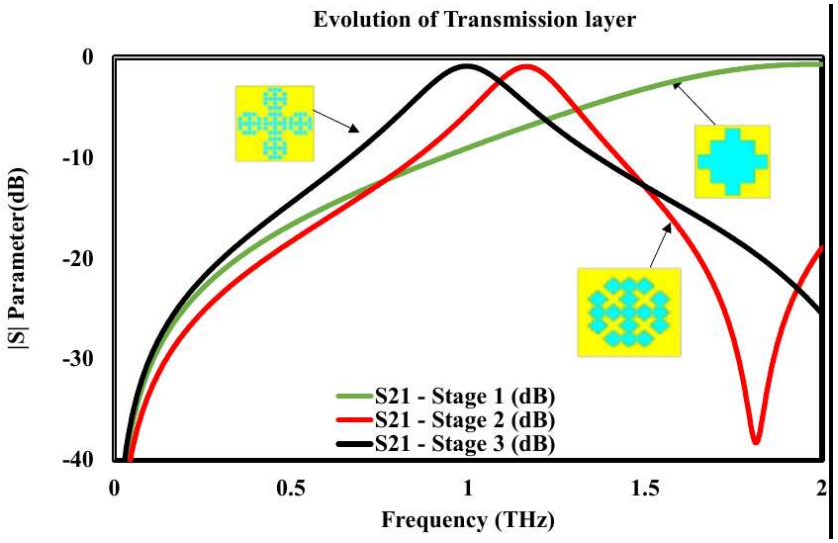


Fig. 3.6 Evolution of Transmission Layer - Simulated S-parameter characteristics

Table 3.3 Summary of the evolution of the transmission layer

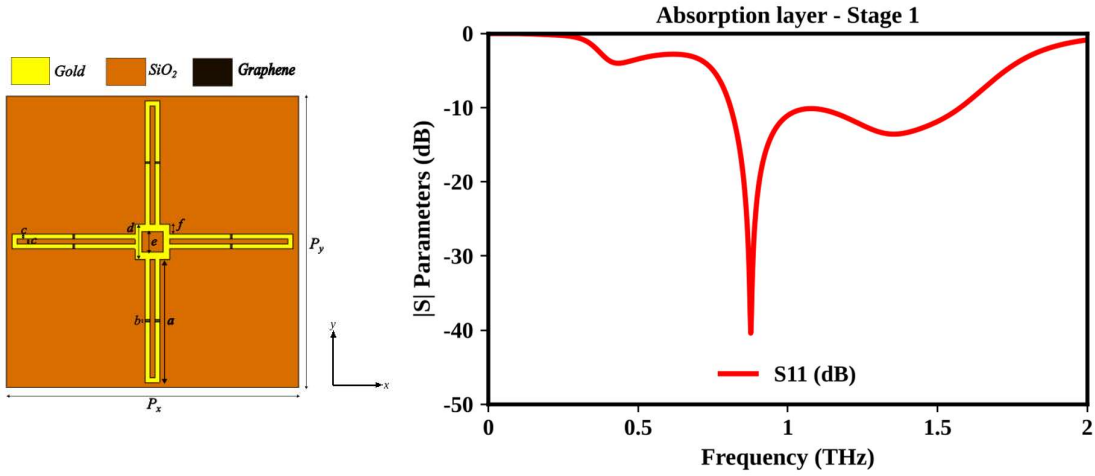
Stage	Unit cell (μm)	Current path (μm)	SiO_2 (μm)	Resonance / peak S21	Outcome
0	120	360	11.5	0.64 THz / −0.51 dB	Basic LC resonance; cell too large
1	60	220	11.5	No distinct passband	Path too short; folding needed
2	60	609.331	11.5	1.17 THz / −0.92 dB	Defined passband; miniaturization shown
3	60	1217.6	4	1.0 THz / −0.87 dB	Target resonance at 1 THz achieved

3.4 DEVELOPMENT AND EVOLUTION OF ABSORPTION LAYER

The absorption (lossy resistive) layer must give a low reflection response over a wide band, so that incident energy outside the passband is absorbed. It is developed in three stages in the same 60 μm × 60 μm cell on a 30 μm SiO_2 substrate with 0.5 μm gold. Absorption is judged from S11: a continuous S11 ≤ −10 dB region indicates that at most 10 % of the incident power is reflected.

3.4.1 Stage 1: Closed-Loop Central Resonator

The initial structure is a central square connected to cross-shaped stems terminated with closed square loops (Fig. 3.7). The closed loops give a limited current path and therefore limited inductive loading, so the response is at relatively high frequencies. There is a sharp resonance of about −40.4 dB at 0.88 THz, but S11 returns above −10 dB around 1 THz, the −10 dB cutoff begins near 0.92 THz and the response extends to about 1.6 THz. The structure cannot reach the desired lower-frequency absorption band, so the loops are modified to lengthen the electrical path.

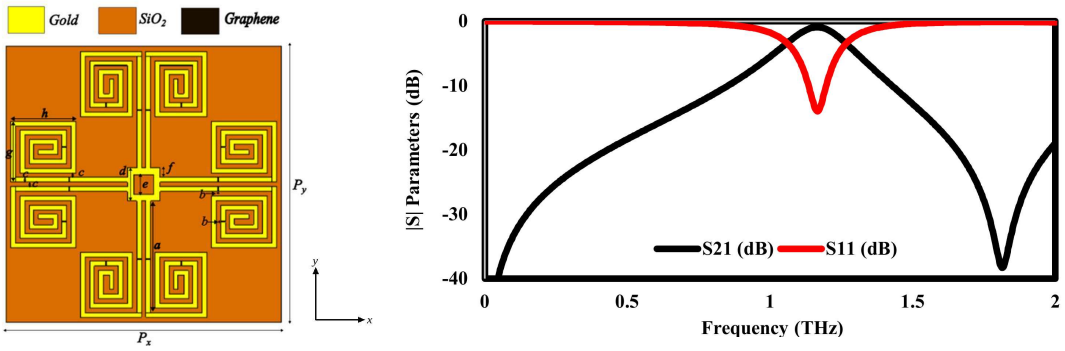


(a) (b)

Fig. 3.7 Structure of the Resistive Layer Stage-1 (a) Front View ($P_x = P_y = 60\mu m$, a = 25.325µm, b = 0.3575µm, c = 1µm, d = 7.15µm, e = 4.29µm, f = 2.075µm) (b) Simulated S-parameter characteristics

3.4.2 Stage 2: Convoluted Square Spiral Structure

In Stage 2 the closed loops are opened and convoluted into square spirals with 1 µm gold width and 1 µm inter-turn gap (Fig. 3.8). The spirals increase the effective current path and add distributed inductive and inter-turn capacitive loading. The response shifts to lower frequencies, with the lower cutoff moving to about 0.81 THz and reflection minima of about −20 dB near 1.04 THz and −18 dB near 1.7 THz. The number of turns that fit in the cell is, however, still insufficient to reach the desired lower-frequency region, so the track width and gap must be reduced.



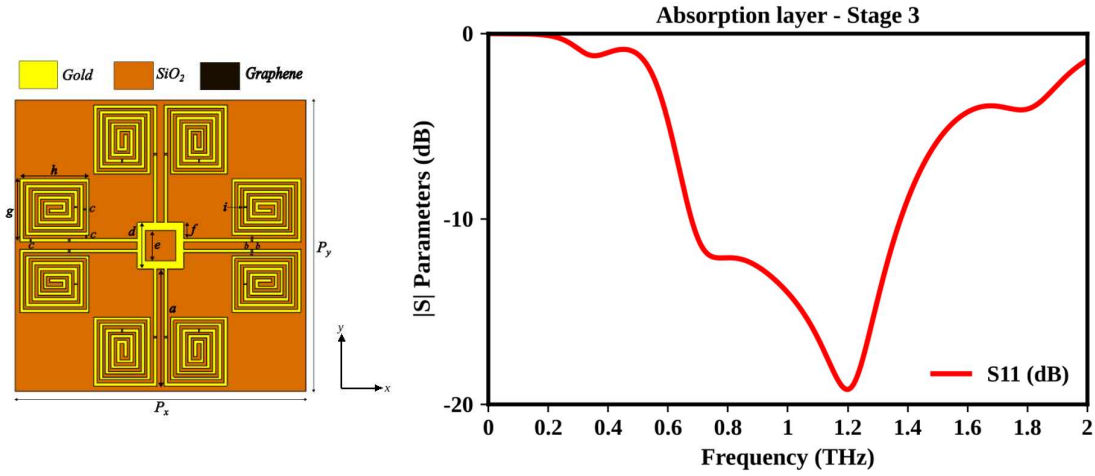
(a) (b)

Fig. 3.8 Structure of the Resistive Layer Stage-2 (a) Front View ($P_x = P_y = 60\mu m$, a = 25.325µm, b = 0.2µm, c = 1µm, d = 7.15µm, e = 4.29µm, f = 2.075µm, g = 13.25µm, h = 14.15µm) (b) Simulated S-parameter characteristics

3.4.3 Stage 3: Graphene-Loaded High-Turn Spiral Structure

In Stage 3 the gold width and inter-turn gap are reduced to 0.6 µm, so that more turns fit inside the same cell and the electrical path is further increased (Fig. 3.9). Graphene (0.001 µm) is introduced on the feeder stem and on the third spiral turn. It adds resistive loss and reactive loading that improve impedance matching. The resulting S11 is continuously below −10 dB from 0.678 THz to 1.377 THz, with a minimum of about −19.2 dB near 1.2 THz. The fractional bandwidth of this band is evaluated in Section 4.2.

The evolution of the absorption layer over the three stages is shown in Fig. 3.10 and summarized in Table 3.4.



(a) (b)

Fig. 3.9 Structure of the Resistive Layer Stage-3 (a) Front View ($P_x = P_y = 60\mu m$, a = 24.15µm, b = 0.3µm, c = 0.6µm, d = 9.5µm, e = 6.29µm, f = 3.25µm, g = 13µm, h = 14.15µm, i = 0.2µm) (b) Simulated S-parameter characteristics

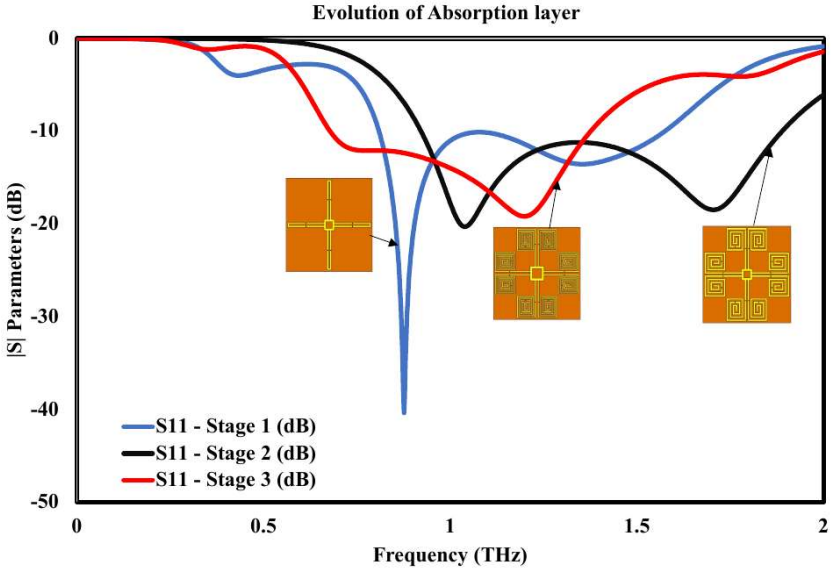


Fig. 3.10 Evolution of Absorption Layer - Simulated S-parameter characteristics

Table 3.4 Summary of the evolution of the absorption layer

Stage	Geometry	Track width / gap	Graphene	Result
1	Closed loops on cross stems	Closed loops	None	−10 dB cutoff ≈ 0.92 THz; no low-frequency band
2	Convoluted square spirals	1 µm / 1 µm	None in the described design	Lower cutoff ≈ 0.81 THz; still too few turns
3	High-turn square spirals	0.6 µm / 0.6 µm	Feeder stem and 3rd turn	S11 ≤ −10 dB over 0.678–1.377 THz

3.5 DEVELOPMENT OF INTEGRATED FSR AND COMBINED SPECTRAL RESPONSE

The optimized transmission and absorption layers are integrated within the same 60 µm × 60 µm unit cell, on one SiO_2 substrate of 30 µm thickness with 0.5 µm gold, to form the single-layer hybrid FSR. Fig. 3.11 shows the absorption pattern on the front face, the side view, and the transmission pattern on the rear face of the single substrate, and the structure is simulated as a whole to see how the transmission resonance and the broadband absorption interact.

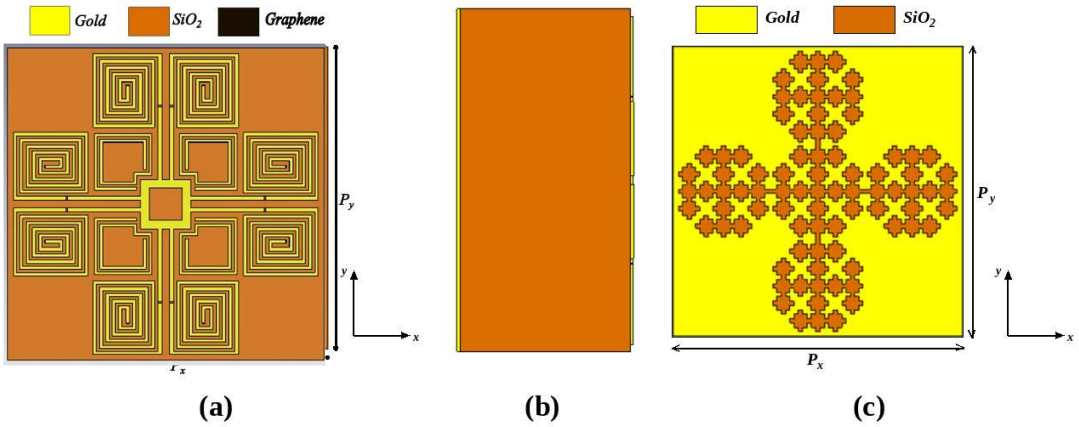


Fig. 3.11 Geometry of the Unit cell. (a) Front view (Absorption), (b) Side view, (c) Rear view (Transmission)

The combined response is shown in Fig. 3.12. The graphene-loaded spiral layer keeps the reflection low over a broad band, with S11 minima of approximately −37 dB near 0.62 THz and −27 dB near 1.33 THz, while S21 shows a broad transmission maximum of approximately −5 dB between about 0.8 THz and 1.0 THz. The transmission resonance of the integrated structure is reported near 0.81 THz. The result demonstrates that frequency-selective transmission and broadband low-reflection absorption can be realised together in one compact unit cell, and it provides the basis for further optimization of the transmission and absorption characteristics.

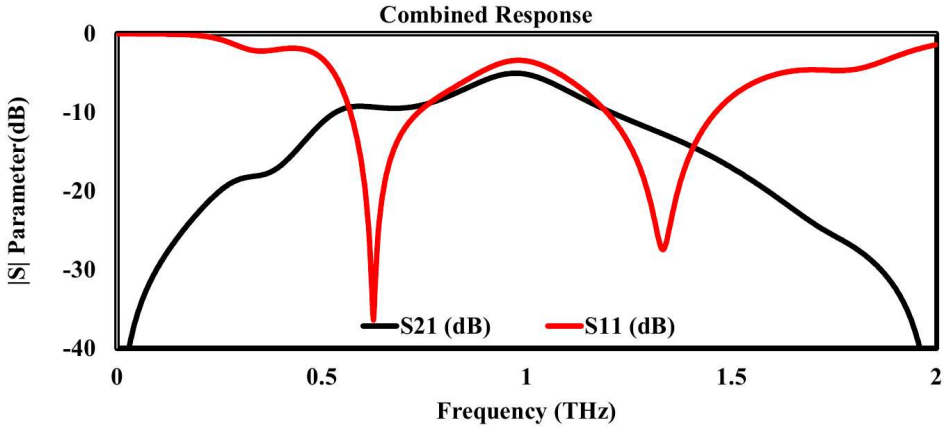


Fig. 3.12 Combined Response - Simulated S-parameter characteristics

CHAPTER 4
RESULTS AND DISCUSSIONS

This chapter discusses the simulated results of the design. The effect of the main design parameters on the response is first summarized in Table 4.1, and the influence of graphene loading, which is the key feature of the absorption layer, is then analysed in detail. The fractional bandwidths of the transmission and absorption layers are calculated next, the angular and polarization stability of the integrated FSR are examined, and the design is then compared with the literature.

4.1 PARAMETRIC ANALYSIS AND IMPACT OF GRAPHENE LOADING

Table 4.1 Effect of the main design parameters on the response

Parameter	Variation studied	Observed effect
Current path of the transmission layer	220 µm → 609.331 µm → 1217.6 µm	Resonance moves from no passband to 1.17 THz and then to 1.0 THz in the same 60 µm cell
Unit cell size	120 µm → 60 µm	Cell halved; $0.2\lambda_0$ at 1 THz, with folding needed to keep resonance
SiO_2 thickness (transmission layer)	11.5 µm → 4 µm	Used together with the longer path to centre the passband at 1 THz
Spiral track width and gap	Closed loops → 1 µm → 0.6 µm	Lower −10 dB cutoff moves from 0.92 → 0.81 → 0.678 THz
Graphene loading	Absent / present; five placements	Isolated resonances become a continuous band; placement controls the response

4.1.1 Performance Analysis With and Without Graphene Loading

Fig. 4.1 compares the S11 of the final spiral structure with and without graphene. Without graphene the response consists of isolated resonances, for example near 0.34 THz and 1.15 THz and smaller ones near 1.5 THz and 1.7 THz, and S11 returns close to 0 dB between them, so the structure reflects much of the incident wave in those regions. When graphene is placed on the feeder

stem and third turn, it contributes resistive loss and reactive loading, which improves impedance matching and broadens the low-reflection response into a single continuous band from 0.678 THz to 1.377 THz. The comparison confirms that graphene is important for obtaining the broadband response.

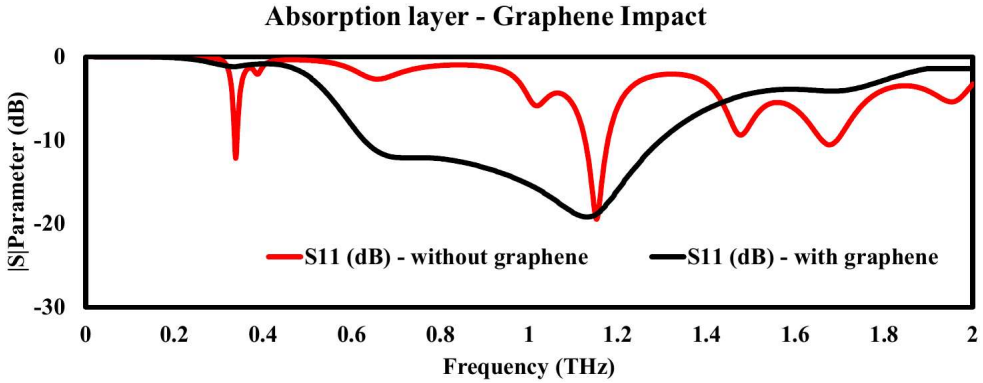


Fig. 4.1 Impact of Graphene on the Absorption Layer - Simulated S11 with and without graphene

4.1.2 Influence of Graphene Placement Across Resonator Locations

The location of the graphene on the spiral was varied over five positions (outer turn, tip, stem only, inner turn and intermediate turn), and the S11 curves are shown in Fig. 4.2. Because current and field are not uniformly distributed along a spiral, each position couples differently with the resonant mode. Graphene on the outer turn gives a very deep but narrow resonance of roughly −50 dB near 1.27 THz, and graphene on the inner turn gives a resonance of about −37 dB near 1.16 THz; both leave regions where the response is above −10 dB. Graphene on the tip or on the stem alone produces moderate resonances of about −20 dB near 1.08 THz and a response that rises above the −10 dB level in between.

The intermediate (third) turn gives the most effective coupling with the resonant mode: the resistive loss and reactive loading it introduces give good impedance matching and a continuous broadband −10 dB response from about 0.678 THz to 1.377 THz. The third-turn location is therefore selected for the final design.

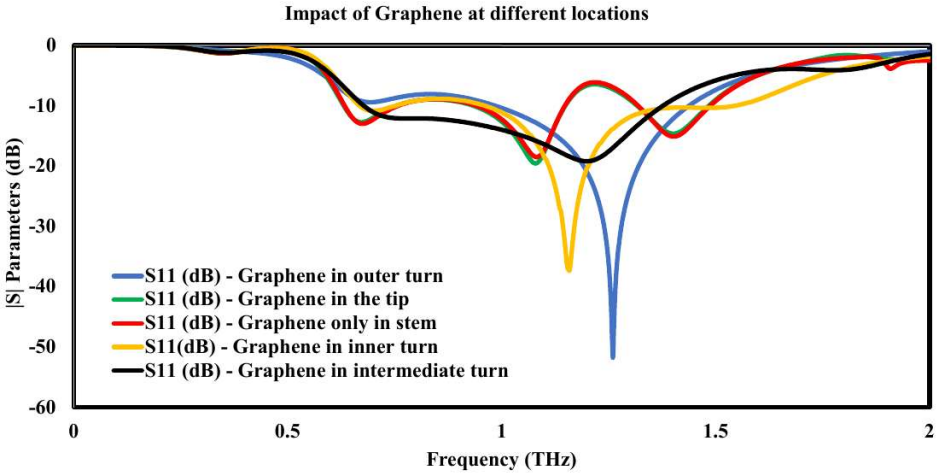


Fig. 4.2 Impact of Graphene Placement at Different Locations on the Spiral - Simulated S11

4.2 FRACTIONAL BANDWIDTH OF TRANSMISSION AND ABSORPTION LAYERS

The fractional bandwidth (FBW) of a band is calculated from its lower and upper band-edge frequencies (f_L and f_H) and the centre frequency (f_c), which is taken here as the 1 THz design frequency:

$$\mathrm{FBW} = \left[\frac{f_H - f_L}{f_c}\right] \times 100\%$$

Transmission layer. For the final bandpass FSS layer (Stage 3), the band edges of the transmission band are 0.88665 THz and 1.1105 THz, so, with the frequencies in THz,

$$\mathrm{FBW_{Tx}} = \left[\frac{1.1105 - 0.88665}{1.0}\right] \times 100\% = 22.385\%$$

Absorption layer. For the graphene-loaded spiral layer, the absorption band is the region in which S11 stays below −10 dB, from 0.6778 THz to 1.3767 THz, so

$$\mathrm{FBW_{Abs}} = \left[\frac{1.3767 - 0.6778}{1.0}\right] \times 100\% = 69.89\%$$

The absorption band is therefore about three times wider than the transmission band (69.89% against 22.385%). This is the intended behaviour of an FSR: a narrow, low-loss transmission window for the wanted channel, and a wide absorption band around it so that unwanted signals on either side are absorbed rather than reflected. The transmission band of 0.887–1.111 THz lies inside the absorption band of 0.678–1.377 THz. These values are for the stand-alone layers; the response of the integrated structure is discussed in Section 3.5 and Section 5.3.

4.3 ANGULAR STABILITY

In practical sub-terahertz links the electromagnetic wave may arrive at different angles of incidence rather than exactly at normal incidence, so the angular stability of the rasorber is an important parameter. To investigate it, the S-parameters and absorptivity of the integrated FSR were simulated for incidence angles of θ = 0°, 15° and 30°, with all geometrical and material parameters kept constant. The results over 0–2 THz are plotted in Fig. 4.3.

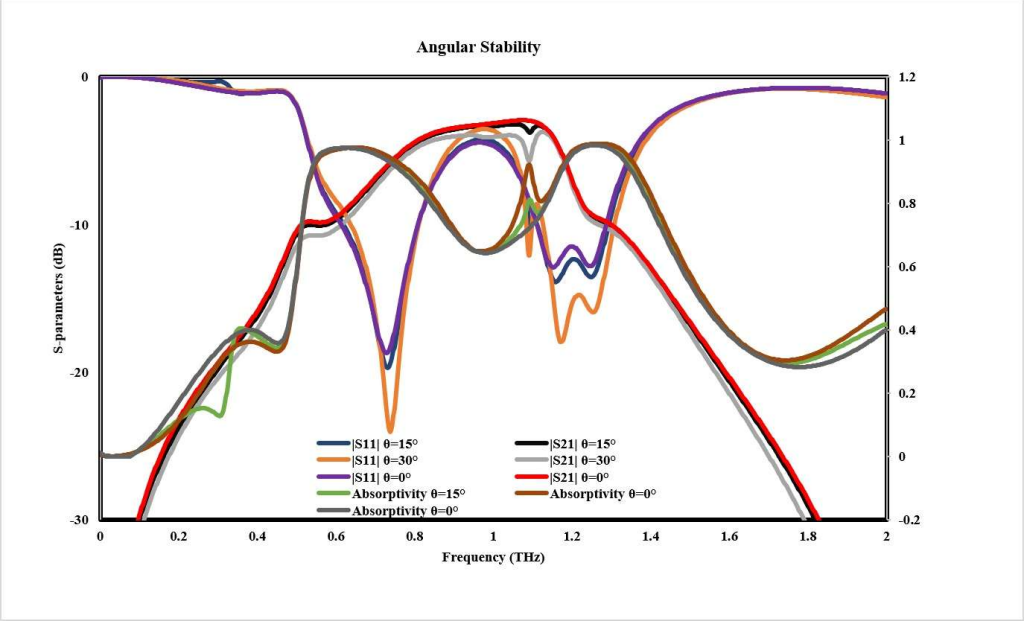


Fig. 4.2 Variation of the S-parameters and absorptivity of the FSR with the angle of incidence θ for TM mode

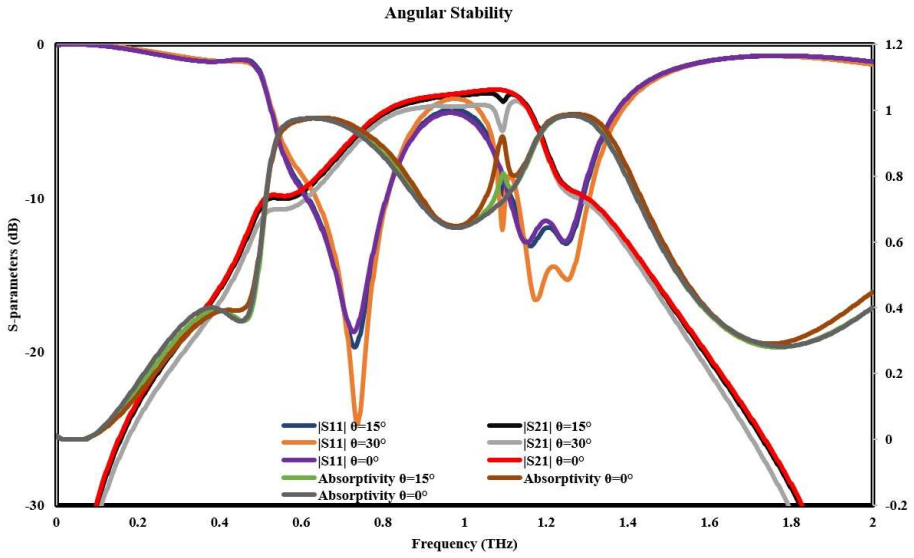


Fig. 4.3 Variation of the S-parameters and absorptivity of the FSR with the angle of incidence θ for TE mode

The key quantities read from Fig. 4.3 are collected in Table 4.2. The values are read from the simulated curves and are therefore approximate. The fractional bandwidth (FBW) of each absorption band is calculated from the band edges at which the absorptivity is 0.9, as FBW = $(f_H - f_L)/f_c \times 100\%$, where f_c is the centre frequency of that band.

Table 4.2 Passband and absorption characteristics of the FSR for different angles of incidence

Parameter	θ = 0°	θ = 15°	θ = 30°
Passband centre frequency	≈ 1.07 THz	≈ 1.07 THz	≈ 1.0 THz (broad)
Peak S21 (insertion loss)	≈ −3.0 dB	≈ −3.0 dB	≈ −3.5 dB
Deepest S11 minimum (lower band)	≈ −18.5 dB at 0.73 THz	≈ −19.7 dB at 0.73 THz	≈ −24.5 dB at 0.74 THz
S11 minima in the upper band	≈ −13 dB near 1.17 THz and 1.26 THz	≈ −13 dB near 1.17 THz and 1.26 THz	≈ −16.5 dB near 1.17 THz and −15 dB near 1.25 THz
Peak absorptivity (lower / upper band)	≈ 0.98 / 0.99	≈ 0.98 / 0.99	≈ 0.98 / 0.99
Lower absorption band (absorptivity ≥ 0.9)	≈ 0.54–0.77 THz	≈ 0.54–0.77 THz	≈ 0.54–0.77 THz
FBW of the lower absorption band	≈ 35.1 %	≈ 35.1 %	≈ 35.1 %
Upper absorption band (absorptivity ≥ 0.9)	≈ 1.18–1.35 THz	≈ 1.18–1.35 THz	≈ 1.18–1.35 THz
FBW of the upper absorption band	≈ 13.4 %	≈ 13.4 %	≈ 13.4 %
Minimum absorptivity between the bands	≈ 0.65 near 0.98 THz	≈ 0.65 near 0.98 THz	≈ 0.65 near 0.98 THz

Passband. At θ = 0° and 15° the transmission coefficient S21 reaches a maximum of about −3.0 dB near 1.07 THz, so the insertion loss of the passband is about 3 dB and its centre frequency does not shift. At 30° the passband becomes a broad plateau between about 0.9 THz and 1.08 THz with a maximum of about −3.5 dB, which is an increase of roughly 0.5 dB in insertion loss. A narrow resonance appears near 1.09 THz for the oblique angles; it is small at 15° and stronger at 30°, where S21 dips to about −6 dB over a very narrow range.

Absorption bands and fractional bandwidth. The absorptivity is above 0.9 from about 0.54 THz to 0.77 THz in the lower band, with a peak of about 0.98 near 0.63 THz, and from about 1.18 THz to 1.35 THz in the upper band, with a peak of about 0.99 near 1.26 THz. Between the two bands it falls to a minimum of about 0.65 near 0.98 THz, which is close to the passband. The

corresponding fractional bandwidths are about 35.1% for the lower band (centre frequency 0.655 THz) and about 13.4% for the upper band (centre frequency 1.265 THz), so the lower band is about 2.6 times wider than the upper band in fractional terms. Within the reading accuracy of the plot, the band edges and peak values are the same for $\theta = 0°$, 15° and 30°, and therefore the two fractional bandwidths are unchanged by the angle of incidence.

Reflection coefficient. The reflection minima keep their positions with the angle. The lower minimum near 0.73 THz deepens from about −18.5 dB at 0° to about −24.5 dB at 30°, and the minima of the upper band deepen from about −13 dB to about −16.5 dB and −15 dB. The reflection is therefore lower, not higher, at the larger angle, which means that the wave is still absorbed rather than reflected.

Discussion. The results show that the proposed FSR has good angular stability up to 30°. The passband stays at about 1.0–1.07 THz with an insertion loss of 3–3.5 dB, the two absorption bands keep the same limits and peak absorptivity, and the fractional bandwidths of about 35.1% and 13.4% are unchanged. The small 60 μm unit cell ($0.2\lambda_0$ at 1 THz) helps to keep the response stable by pushing grating effects away from the operating band. The main changes are the slightly higher insertion loss and the narrow resonance near 1.09 THz at 30°. Angles beyond 30° were not simulated and are left for future work.

4.4 POLARIZATION STABILITY

In practical sub-terahertz links the polarization of the incident wave is not always known or fixed, so a rasorber should keep its passband and absorption bands unchanged when the polarization changes. To examine this, the S-parameters of the integrated FSR were simulated at normal incidence ($\theta = 0°$) for TE polarization ($\varphi = 0°$) and TM polarization ($\varphi = 90°$), in which the electric

field is along two orthogonal directions in the plane of the unit cell, with all geometrical and material parameters kept constant. The simulated |S11|, |S21| and absorptivity are plotted over 0–2 THz in Fig. 4.4.

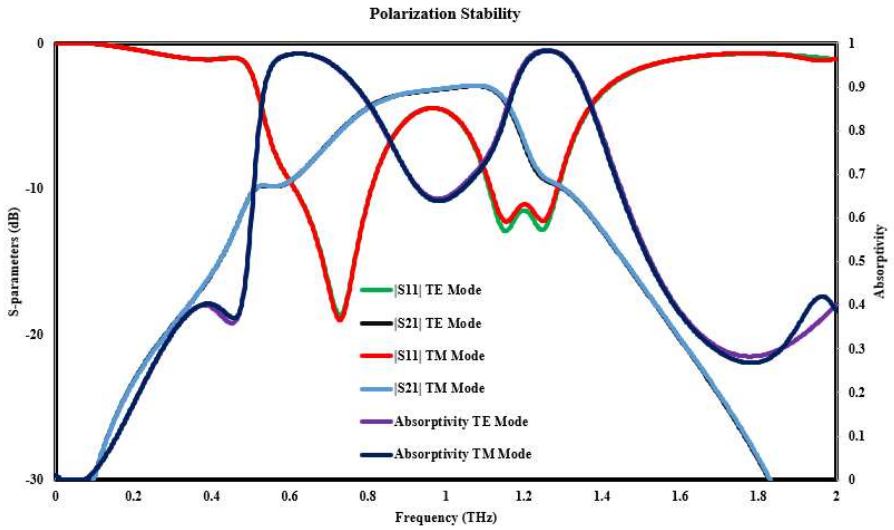


Fig. 4.4 Variation of the S-parameters and absorptivity of the FSR for TE (θ = 0°, φ = 0°) and TM (θ = 0°, φ = 90°) polarizations

The key quantities read from Fig. 4.4 are collected in Table 4.3. The values are read from the simulated curves and are therefore approximate.

Table 4.3 Passband and absorption characteristics of the FSR for TE and TM polarizations

Parameter	TE polarization	TM polarization
Passband centre frequency	≈ 1.07 THz	≈ 1.07 THz
Peak S21 (insertion loss)	≈ −2.95 dB	≈ −2.95 dB
Deepest S11 minimum	≈ −19 dB at 0.73 THz	≈ −19 dB at 0.73 THz
S11 minima in the upper band	≈ −12.9 dB and −12 dB near 1.15 THz and 1.25 THz	≈ −12.3 dB and −12 dB near 1.15 THz and 1.25 THz
Peak absorptivity	≈ 0.98 at 0.63 THz and ≈ 0.98 at 1.25 THz	≈ 0.98 at 0.63 THz and ≈ 0.98 at 1.25 THz
Absorptivity ≥ 0.9	≈ 0.54–0.77 THz and ≈ 1.18–1.35 THz	≈ 0.54–0.77 THz and ≈ 1.18–1.35 THz
Fractional bandwidth of the lower absorption band	≈ 35.1 %	≈ 35.1 %
Fractional bandwidth of the upper absorption band	≈ 13.4 %	≈ 13.4 %
Minimum absorptivity between the bands	≈ 0.64 near 0.99 THz	≈ 0.64 near 0.99 THz

Passband. The TE and TM curves overlap almost exactly over the whole 0–2 THz range. The transmission coefficient S21 reaches a maximum of about −2.95 dB near 1.07 THz for both polarizations, so the insertion loss of the passband is about 3 dB and its centre frequency does not shift with polarization. S21 stays within about 0.8 dB of this maximum from roughly 0.87 THz to 1.15 THz, and the passband lies between the two absorption bands, as required for an ATA response.

Absorption bands. The absorptivity rises steeply from about 0.4 near 0.45 THz to above 0.9 at about 0.54 THz and reaches a peak of about 0.98 near 0.63 THz. It stays above 0.9 up to about 0.77 THz and then falls to a minimum of about 0.64 near 0.99 THz, close to the passband. A second absorption band appears above the passband, with absorptivity above 0.9 from about 1.18 THz to 1.35 THz and a peak of about 0.98 near 1.25 THz. The fractional bandwidth of the lower absorption band (0.54–0.77 THz) is about 35.1% and that of the upper band (1.18–1.35 THz) is about 13.4%, each calculated about the centre frequency of its own band. Beyond 1.4 THz the absorptivity decreases and the

transmission falls steadily, reaching −30 dB near 1.83 THz. The reflection coefficient is below −10 dB around the two absorption peaks, with a deepest minimum of about −19 dB near 0.73 THz and minima of about −12 dB near 1.15 THz and 1.25 THz.

TE and TM comparison. The only visible differences between the two polarizations are small. The S11 minimum near 1.15 THz differs by about 0.5 dB, and the absorptivity differs by up to about 0.03 near 0.45 THz and between 1.8 THz and 2 THz. These regions lie outside the passband and the main absorption peaks, so the frequency, depth and width of the bands are not affected.

Discussion. The results show that the proposed FSR is polarization stable at normal incidence. The passband stays at about 1.07 THz with an insertion loss of about 3 dB, and the lower and upper absorption bands keep the same limits and peak values for both polarizations. This behaviour can be attributed to the small $0.2\lambda_0$ unit cell and the near-symmetric layout of the folded fractal transmission pattern, which let the structure respond almost equally to the two orthogonal field directions. The FSR is therefore suitable for sub-terahertz applications in which the polarization of the incident wave may change. Only the two orthogonal polarizations at normal incidence were simulated here; the effect of the angle of incidence is discussed in Section 4.3.

4.5 COMPARATIVE ANALYSIS WITH LITERATURE

Table 4.4 compares the proposed design with the reviewed FSRs. The electrical size is computed as $P \cdot f/c$, where P is the larger side of the unit cell, f is the transmission frequency (the centre frequency for a band, or the higher of two bands for dual-band devices) and c is the speed of light.

Table 4.4 Comparison of the proposed FSR with reported terahertz designs

Design	Period (μm)	Transmission	Electrical size	Insertion loss	Absorption band	Passive
[3] 2025	20	3 THz	0.20λ₀	2.6 dB	1–10 THz	No (optical)
[7] 2024	160	1.053 THz	0.56λ₀	≤ 0.46 dB	0.44–2.45 THz (discrete)	No (VO_2)
[6] 2022	150	0.90 THz	0.45λ₀	0.87 dB	0.66–2.48 THz (discrete)	No (VO_2)
[5] 2024	35	5.3–10 THz	0.89λ₀	0.65 dB	3.6–9.4 THz	No (VO_2)
[2] 2024	24	1–10 THz	0.44λ₀	< 0.46 dB	2.54–7.65 THz	No (VO_2)
[4] 2023	14	5.15 THz	0.24λ₀	4.29 dB	3.31–10 THz	No (thermal)
This work	**60**	**1.0 THz**	**0.20λ₀**	-2.91dB	**0.678–1.377 THz**	**Yes**

The proposed design is the only one in the comparison that operates in the sub-THz range with a cell of 0.2λ₀ and no switching stimulus. Its unit cell is as electrically small as the smallest reported design [3], whose operating frequency is three times higher, and is much smaller than the designs of [6] and [7] that work in a comparable frequency range. The insertion loss of the bandpass layer alone (0.87 dB) is comparable with that reported for [6] and lower than that of [3] and [4]; the loss of the integrated structure is higher, as discussed in Section 5.3.

4.6 KEY MERITS AND PERFORMANCE ADVANTAGES OF PROPOSED FSR DESIGN

- **Compact unit cell.** The 60 μm × 60 μm cell corresponds to 0.2λ₀ at 1 THz, below the 0.3λ₀ target for sub-wavelength metasurfaces and favourable to angular stability.
- **Electrical miniaturization by path folding.** A current path of 1217.6 μm is fitted inside the 60 μm cell by fractal folding, about twenty times the cell side.
- **Low-loss transmission layer.** The bandpass FSS layer has a peak S21 of −0.87 dB at 1 THz with a fractional bandwidth of 22.385%.

- **Wideband absorption.** The graphene-loaded spiral layer keeps S11 below −10 dB from 0.678 THz to 1.377 THz, a fractional bandwidth of 69.89%.
- **Passive and lumped-element-free.** The design uses SiO_2, gold and graphene only. It needs no lumped resistors, no active devices and no switching stimulus.
- **Single-layer, low-profile topology.** Transmission and absorption are realised on a single substrate with no air spacer, which keeps the structure thin.
- **Stable under oblique incidence.** For incidence angles of 0°, 15° and 30° the passband stays at about 1.0–1.07 THz with an insertion loss of 3–3.5 dB, and the lower and upper absorption bands, with fractional bandwidths of about 35.1% and 13.4%, keep their limits.
- **Polarization-stable response.** The TE and TM responses at normal incidence overlap almost exactly, with the passband at about 1.07 THz and absorptivity peaks of about 0.98 unchanged by polarization.
- **Targeted at 6G and sensing bands.** The 0–2 THz operating range matches the sub-THz bands considered for 6G links, THz sensing and imaging.

CHAPTER 5
CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

This project presented the design of a single-layer hybrid miniaturized frequency selective rasorber for sub-terahertz applications. A 60 μm × 60 μm unit cell ($0.2\lambda_0$ at 1 THz) on a SiO_2 substrate with gold patterns was developed in CST Studio Suite in two parts. The bandpass FSS layer was evolved from a 120 μm square loop resonating at 0.64 THz, through a cross resonator, to a folded fractal geometry whose 1217.6 μm current path gives a transmission band at 1 THz with a peak S21 of −0.87 dB and a fractional bandwidth of 22.385%. The resistive layer was evolved from closed loops, through convoluted spirals, to a 0.6 μm track-width spiral structure with graphene on the feeder stem and third turn, which gives a continuous −10 dB reflection response from 0.678 THz to 1.377 THz (69.89%). Parametric studies showed how current-path length, track width and gap, substrate thickness and graphene placement shape the response, and the third-turn graphene location was found to give the best impedance matching. The two layers were integrated in a single cell, and the combined response confirmed the feasibility of the hybrid concept. Simulations at incidence angles of 0°, 15° and 30° show that the passband stays at about 1.0–1.07 THz with an insertion loss of 3–3.5 dB and that the lower and upper absorption bands, with fractional bandwidths of about 35.1% and 13.4%, are retained. The simulated TE and TM responses of the integrated FSR at normal incidence overlap almost exactly, which shows that the response is stable with polarization.

5.2 MAJOR RESEARCH CONTRIBUTIONS

- A passive, single-layer hybrid FSR concept for the 0–2 THz band with a unit cell of $0.2\lambda_0$, filling a gap in the reviewed literature, which concentrates on frequencies above 2.5 THz or on switchable devices.
- A stage-wise miniaturization procedure for the bandpass layer that shows, with simulated evidence at every stage, how current-path length controls the resonant frequency in a fixed cell.
- A graphene-loaded spiral resistive layer that avoids lumped resistors and gives a 69.89% fractional bandwidth with S11 ≤ −10 dB.
- A systematic study of graphene placement on the spiral, identifying the intermediate turn as the most effective position.

5.3 LIMITATIONS OF PRESENT WORK

- The results are from simulation only. The structure has not been fabricated or measured.
- In the integrated structure the transmission maximum is lower (about −5 dB) than that of the stand-alone bandpass layer (−0.87 dB), and the transmission resonance is reported near 0.81 THz rather than at the 1 THz design frequency. The coupling between the two patterns therefore needs further optimization.
- The substrate thickness differs between the stand-alone transmission layer (4 μm) and the integrated structure (30 μm), and the effect of this on the passband has not been separated out.
- Angular stability was examined only up to $\theta = 30°$ (Section 4.3), and polarization dependence only for TE and TM excitation at normal incidence (Section 4.4). Absorptivity of the stand-alone layers was assessed from S11 only.

- The graphene is modelled as a thin layer of fixed properties, and the sensitivity of the response to graphene quality and fabrication tolerances has not been studied.

5.4 FUTURE SCOPE AND RECOMMENDATIONS

- Optimize the integrated structure so that the passband returns to 1 THz with an insertion loss close to that of the stand-alone FSS layer, for example by tuning the substrate thickness and the spacing between the two patterns.
- Study incidence angles beyond 30°, and polarization at oblique angles, to confirm angular stability, and calculate RCS reduction.
- Use the simulated dataset from further parametric sweeps to develop an AI or machine-learning model that predicts FSR dimensions for a target frequency and reduces design time.
- Explore electrically tunable graphene, through gating, to make the absorption band adjustable.
- Fabricate the structure and verify it by THz time-domain spectroscopy measurements.
- Extend the design to other sub-THz bands proposed for 6G.

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